

CHIAYI, Taiwan

WMO: 467480

Lat: 23.4959N

Lon: 120.4329E

Elev: 27

StdP: 101.00

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.3	10.7	4.9	5.4	10.8	6.7	6.1	12.2	7.4	17.1	6.9	16.0	2.3	10	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	33.6	26.9	33.0	26.9	32.4	26.7	27.7	32.2	27.4	31.8	27.1	31.5	4.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	22.1	30.8	26.1	21.6	30.3	25.8	21.1	29.9	88.9	32.5	87.1	32.0	85.5	31.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.7	5.9	5.3	6.3	35.0	2.0	0.4	4.9	35.3	3.7	35.6	2.6	35.8	1.2	36.1
			5.5	28.6	1.8	0.3	4.2	28.9	3.2	29.1	2.2	29.3	0.9	29.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.6	17.1	17.6	20.3	23.3	25.9	28.1	28.9	28.5	27.2	24.7	22.4	19.3	(6)
(7)		DBStd	4.66	2.55	2.79	3.16	3.18	1.89	1.59	1.11	1.06	1.46	1.96	2.18	2.55	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	144	54	45	18	6	0	0	0	0	0	0	2	18	(9)
(10)		CDD10.0	4975	219	212	319	398	494	543	584	572	517	456	372	288	(10)
(11)		CDD18.3	2076	15	24	79	154	236	293	326	314	267	197	123	48	(11)
(12)		CDH23.3	18691	44	85	365	1054	2002	3242	3863	3521	2543	1264	562	146	(12)
(13)		CDH26.7	6486	3	8	70	281	649	1289	1621	1311	828	316	96	15	(13)
(14)	Wind	WSAvg	2.4	2.8	2.7	2.5	2.3	2.1	2.5	2.5	2.4	2.1	2.1	2.1	2.3	(14)
(15)	Precipitation	PrecAvg	1967	27	52	55	110	172	369	420	438	223	49	27	24	(15)
(16)		PrecMax	3230	90	328	159	457	406	916	960	1168	1026	327	211	95	(16)
(17)		PrecMin	1028	0	0	4	0	9	23	61	98	41	0	0	0	(17)
(18)		PrecStd	565	22	72	48	105	98	225	240	304	231	74	45	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.6	27.9	30.5	31.7	32.9	34.6	34.2	33.7	32.8	31.5	30.2	28.6	(19)
(20)			MCWB	19.2	21.7	22.8	25.3	27.0	26.7	27.1	27.5	27.1	26.1	23.6	22.1	(20)
(21)		2%	DB	24.4	25.9	28.6	30.6	31.9	33.6	33.6	33.0	32.1	30.5	29.0	26.4	(21)
(22)			MCWB	18.7	20.5	22.8	24.7	26.3	26.7	26.9	27.3	26.8	25.5	23.1	20.7	(22)
(23)		5%	DB	23.0	24.1	26.9	29.6	31.0	32.9	33.0	32.3	31.4	29.6	27.8	24.9	(23)
(24)			MCWB	18.1	19.6	22.0	24.2	25.9	26.7	26.8	27.0	26.5	24.8	22.5	19.7	(24)
(25)		10%	DB	21.4	22.4	25.4	28.5	30.1	32.1	32.4	31.7	30.7	28.6	26.6	23.5	(25)
(26)			MCWB	17.3	18.7	21.1	23.8	25.4	26.4	26.7	26.8	26.2	24.2	21.7	18.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	22.6	24.1	26.0	27.7	28.0	28.1	28.3	27.8	26.7	24.6	23.1	(27)
(28)			MCDWB	24.6	27.0	28.9	30.7	31.9	33.0	33.1	32.6	31.7	30.6	28.8	27.9	(28)
(29)		2%	WB	19.2	21.3	23.2	25.2	26.8	27.3	27.6	27.7	27.3	25.9	23.7	21.5	(29)
(30)			MCDWB	23.5	24.9	27.5	29.8	31.0	32.3	32.5	32.0	31.1	29.5	27.9	24.9	(30)
(31)		5%	WB	18.5	20.1	22.4	24.6	26.2	26.9	27.2	27.3	26.9	25.3	23.0	20.6	(31)
(32)			MCDWB	22.2	23.3	26.2	28.9	30.4	31.8	32.0	31.5	30.6	28.6	26.8	23.5	(32)
(33)		10%	WB	17.6	18.9	21.5	24.0	25.6	26.5	26.8	26.9	26.4	24.7	22.2	19.6	(33)
(34)			MCDWB	20.7	22.4	24.9	27.9	29.5	31.1	31.4	30.9	29.9	27.7	25.5	22.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.5	7.5	8.0	7.7	7.5	7.0	7.1	6.6	7.2	8.1	8.9	9.1	(35)
(36)			MCDBR	10.6	10.0	9.9	9.1	8.5	8.5	8.0	7.6	7.9	8.6	9.7	10.3	(36)
(37)			MCWBR	5.6	5.3	5.1	4.7	4.2	3.6	3.4	3.4	3.6	4.2	4.6	5.2	(37)
(38)		5% WB	MCDWB	9.4	9.1	9.5	8.5	8.2	8.0	7.5	7.3	7.4	7.7	8.7	8.7	(38)
(39)			MCWBR	5.1	5.3	5.2	4.6	4.3	3.7	3.5	3.5	3.6	3.9	4.4	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.509	0.564	0.694	0.698	0.590	0.534	0.544	0.602	0.631	0.612	0.533	0.522	(40)	
(41)		taud	1.965	1.861	1.612	1.655	1.923	2.093	2.056	1.884	1.792	1.791	1.968	1.952	(41)	
(42)		Ebn at Noon	739	727	656	667	739	776	768	725	692	678	712	707	(42)	
(43)		Edh at Noon	166	196	263	256	195	163	169	201	216	207	164	163	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.41	3.82	4.13	4.41	4.78	5.04	5.49	4.95	4.83	4.52	3.60	3.32	(44)	
(45)		RadStd	0.37	0.47	0.44	0.55	0.59	0.53	0.40	0.51	0.38	0.28	0.29	0.33	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.28	N/A	N/A	N/A	+0.18	+0.25	N/A	N/A	+111

Nomenclature: See separate page